

Experiment - 04

AIM:-

1. Display the list of employees who have joined the company before 30th June 80 or after 31st Dec 81.
2. Display the names of employees whose names have second alphabet A in their names.
3. Display the names of employees whose name is exactly five characters in length.
4. Display the names of employees whose names have second alphabet A in their names.
5. Display the names of employees who are not working as salesman or clerk or analyst.
6. Display the name of the employee along with their annual salary (sal*12). The name of the employee earning highest salary should appear first.
7. Display name, sal, hra, pf, da, totalsal for each employee. The output should be in the order of total sal, hra 15% of sal, da 10% of sal, pf 5% of sal. Total salary will be (sal*hra*da)-pf.
8. Update the salary of each employee by 10% increment who are not eligible for commission.
9. Display those employees whose salary is more than 3000 after giving 20% increment.
10. Display those employees whose salary contains atleast 3 digits.

PROCEDURE:-

- **SELECT** – Retrieve data
- **WHERE** – Filter records
- **LIKE** – Pattern matching
- **IN / NOT IN** – Multiple values condition
- **ORDER BY** – Sort data
- **UPDATE** – Modify records
- **IS NULL / IS NOT NULL** – Check null values
- **Arithmetic Operators (+, *)** – Perform calculations

Perform Following Query using Employee Table:-

1. Display the list of employees who have joined the company before 30th June 80 or after 31st

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Dec 81.

Ans-

SELECT ename, hiredate FROM Employee WHERE hiredate < '1980-06-30' OR hiredate > '1981-12-31';

```
MariaDB [2cse25_1650]> SELECT ename, hiredate
-> FROM Employee
-> WHERE hiredate < '1980-06-30'
-> OR hiredate > '1981-12-31';
+-----+-----+
| ename  | hiredate |
+-----+-----+
| SCOTT  | 1982-12-09 |
| ADAMS  | 1983-01-12 |
| MILLER | 1982-01-23 |
+-----+-----+
3 rows in set (0.003 sec)
```

2. Display the names of employees whose names have second alphabet A in their names.

Ans-

SELECT ename FROM Employee WHERE ename LIKE '_A%';

```
MariaDB [2cse25_1650]> SELECT ename
-> FROM Employee
-> WHERE ename LIKE '_A%';
+-----+
| ename |
+-----+
| WARD  |
| MARTIN |
| JAMES |
+-----+
3 rows in set (0.001 sec)
```

3. Display the names of employees whose name is exactly five characters in length.

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Ans-

```
SELECT ename FROM Employee WHERE ename LIKE '_____';
```

```
MariaDB [2cse25_1650]> SELECT ename
-> FROM Employee
-> WHERE ename LIKE '_____';
+-----+
| ename |
+-----+
| SMITH |
| ALLEN |
| JONES |
| BLAKE |
| CLARK |
| SCOTT |
| ADAMS |
| JAMES |
+-----+
8 rows in set (0.001 sec)
```

4. Display the names of employees whose names have second alphabet A in their names.

Ans-

```
SELECT ename FROM Employee WHERE ename LIKE '_A%';
```

```
MariaDB [2cse25_1650]> SELECT ename
-> FROM Employee
-> WHERE ename LIKE '_A%';
+-----+
| ename |
+-----+
| WARD   |
| MARTIN |
| JAMES  |
+-----+
3 rows in set (0.001 sec)
```

5. Display the names of employees who are not working as salesman or clerk or analyst.

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Ans-

```
SELECT ename FROM Employee WHERE job NOT IN ('SALESMAN','CLERK','ANALYST');
```

```
MariaDB [2cse25_1650]> SELECT ename
-> FROM Employee
-> WHERE job NOT IN ('SALESMAN','CLERK','ANALYST');
+-----+
| ename |
+-----+
| JONES |
| BLAKE |
| CLARK |
| KING  |
+-----+
4 rows in set (0.001 sec)
```

6. Display the name of the employee along with their annual salary (sal*12). The name of the employee earning highest salary should appear first.

Ans-

```
SELECT ename, sal*12 AS annual_salary FROM Employee ORDER BY sal DESC;
```

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```
MariaDB [2cse25_1650]> SELECT ename, sal*12 AS annual_salary  
-> FROM Employee  
-> ORDER BY sal DESC;
```

ename	annual_salary
KING	66000.00
SCOTT	39600.00
FORD	39600.00
JONES	39270.00
BLAKE	37620.00
CLARK	32340.00
ALLEN	19200.00
TURNER	18000.00
MILLER	17160.00
MARTIN	15000.00
WARD	15000.00
ADAMS	14520.00
JAMES	12540.00
SMITH	10560.00

```
14 rows in set (0.001 sec)
```

7. Display name, sal, hra, pf, da, totalsal for each employee. The output should be in the order of total sal, hra 15% of sal, da 10% of sal, pf 5% of sal. Total salary will be (sal*hra*da)-pf.

Ans-

```
SELECT ename, sal, sal*0.15 AS hra, sal*0.10 AS da, sal*0.05 AS pf, (sal + sal*0.15 + sal*0.10 -  
sal*0.05) AS totalsal FROM Employee ORDER BY totalsal;
```

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```
MariaDB [2cse25_1650]> SELECT ename, sal,  
-> sal*0.15 AS hra,  
-> sal*0.10 AS da,  
-> sal*0.05 AS pf,  
-> (sal + sal*0.15 + sal*0.10 - sal*0.05) AS totalsal  
-> FROM Employee  
-> ORDER BY totalsal;
```

ename	sal	hra	da	pf	totalsal
SMITH	880.00	132.0000	88.0000	44.0000	1056.0000
JAMES	1045.00	156.7500	104.5000	52.2500	1254.0000
ADAMS	1210.00	181.5000	121.0000	60.5000	1452.0000
WARD	1250.00	187.5000	125.0000	62.5000	1500.0000
MARTIN	1250.00	187.5000	125.0000	62.5000	1500.0000
MILLER	1430.00	214.5000	143.0000	71.5000	1716.0000
TURNER	1500.00	225.0000	150.0000	75.0000	1800.0000
ALLEN	1600.00	240.0000	160.0000	80.0000	1920.0000
CLARK	2695.00	404.2500	269.5000	134.7500	3234.0000
BLAKE	3135.00	470.2500	313.5000	156.7500	3762.0000
JONES	3272.50	490.8750	327.2500	163.6250	3927.0000
FORD	3300.00	495.0000	330.0000	165.0000	3960.0000
SCOTT	3300.00	495.0000	330.0000	165.0000	3960.0000
KING	5500.00	825.0000	550.0000	275.0000	6600.0000

14 rows in set (0.001 sec)

8. Update the salary of each employee by 10% increment who are not eligible for commission.

Ans-

UPDATE Employee SET sal = sal * 1.10 WHERE comm IS NULL;

```
MariaDB [2cse25_1650]> UPDATE Employee  
-> SET sal = sal * 1.10  
-> WHERE comm IS NULL;  
Query OK, 10 rows affected (0.009 sec)  
Rows matched: 10 Changed: 10 Warnings: 0
```

9. Display those employees whose salary is more than 3000 after giving 20% increment.

Ans-

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SELECT ename FROM Employee WHERE sal*1.20 > 3000;

```
MariaDB [2cse25_1650]> SELECT ename
-> FROM Employee
-> WHERE sal*1.20 > 3000;
+-----+
| ename |
+-----+
| JONES |
| BLAKE |
| CLARK |
| SCOTT |
| KING  |
| FORD  |
+-----+
6 rows in set (0.002 sec)
```

10. Display those employees whose salary contains atleast 3 digits.

Ans-

SELECT ename, sal FROM Employee WHERE sal >= 100;

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```
MariaDB [2cse25_1650]> SELECT ename, sal  
-> FROM Employee  
-> WHERE sal >= 100;
```

ename	sal
SMITH	968.00
ALLEN	1600.00
WARD	1250.00
JONES	3599.75
MARTIN	1250.00
BLAKE	3448.50
CLARK	2964.50
SCOTT	3630.00
KING	6050.00
TURNER	1500.00
ADAMS	1331.00
JAMES	1149.50
FORD	3630.00
MILLER	1573.00

```
14 rows in set (0.001 sec)
```